I am going to build a AI model like deepseek and openai (gpt, o models).
From which books do I have to start reading to kick in?

If you're aiming to build models in the same league as **OpenAI GPT/o-series** or **DeepSeek**, you're not just learning "ML"—you're stacking **math → ML → deep learning → transformers → systems → scaling**.

I'll give you a **serious, no-fluff reading path** (books + why each matters), ordered so you don't waste time.

🧠 1. Foundations (Math + Intuition)

You *can't skip this* if you want to build models, not just use them.

📘 Must-read

- **Mathematics for Machine Learning – Deisenroth**
  - Covers linear algebra, probability, optimization (the exact math used in ML papers)
- **Pattern Recognition and Machine Learning – Christopher Bishop**
  - Classic. Dense but gives deep probabilistic understanding
- **The Elements of Statistical Learning – Hastie, Tibshirani**
  - Theoretical backbone of ML (still referenced today)

👉 If you struggle with math, add:

- **Linear Algebra Done Right – Axler**

🤖 2. Core Machine Learning

Before deep learning, understand *why models work*.

📘 Must-read

- **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron**
  - Practical + builds intuition fast
- **Machine Learning: A Probabilistic Perspective – Kevin Murphy**
  - One of the most complete ML books ever written

🔥 3. Deep Learning (Critical Layer)

This is where modern AI begins.

📘 Core book

- **Deep Learning**
  - The "Bible" of deep learning
  - Covers neural nets, backprop, optimization

👉 You should *fully understand*:

- Backpropagation
- Gradient descent variants
- CNNs / RNNs basics

⚡ 4. Transformers & LLMs (Your Actual Goal)

This is the **heart of GPT/DeepSeek-like systems**.

📘 Must-read

- **Natural Language Processing with Transformers – Lewis et al.**
  - Practical transformer implementation
- **Speech and Language Processing – Jurafsky & Martin**
  - Classic NLP reference

📄 Papers (more important than books here)

You *must* read:

- **Attention Is All You Need**
- **GPT-3 Paper**
- **LLaMA Paper**

👉 This is where you *actually learn how GPT works*

🏗️ 5. Systems & Scaling (What Most People Ignore)

This is what separates hobby projects from **OpenAI-level systems**.

📘 Read

- **Designing Data-Intensive Applications – Martin Kleppmann**
  - Distributed systems, data pipelines, scaling
- **Machine Learning Systems Design – Chip Huyen**
  - Production ML systems

⚙️ 6. Optimization, Training, and Efficiency

This is how you actually *train large models*.

📘 Recommended

- **Deep Learning for Coders with fastai and PyTorch – Jeremy Howard**
  - Practical training tricks

🧩 Reality Check (Important)

Books alone will NOT get you there.

To reach **DeepSeek/OpenAI level**, you must also:

- Re-implement:
  - Transformer from scratch
  - GPT-style training loop
- Train small LLMs (100M–1B params)
- Learn:
  - Distributed training (DDP, FSDP)
  - CUDA / GPU optimization
  - Tokenization (BPE, SentencePiece)
- Study open models:
  - LLaMA
  - Mistral
  - DeepSeek

🧭 Suggested Order (Efficient Path)

If you want a **fast but solid route**:

1. Mathematics for ML
2. Hands-On ML
3. Deep Learning (Goodfellow)
4. Transformers book
5. Papers (Attention → GPT → LLaMA)
6. Systems design books

🚀 If your goal is *VERY serious* (like founding a model company)

Say so—I can give you a **0→LLM roadmap** including:

- exact projects (build your own GPT)
- compute requirements
- dataset strategies
- architecture decisions (Mixture-of-Experts, etc.)

That's a different level from just "reading books."